

Clinical Treatment-Outcome Prediction: Model Comparison & API Recommendation

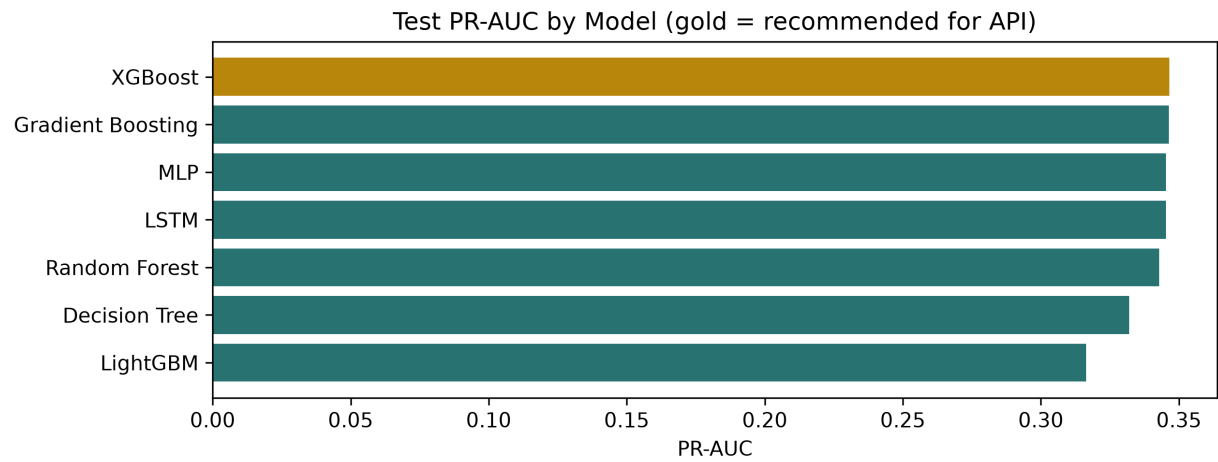
Generated 2026-09-08 11:45 — 7 models trained on 1,015,405 patient records, leakage-safe 60/20/20 split, primary metric = PR-AUC (imbalanced target: ~22.3% Effective).

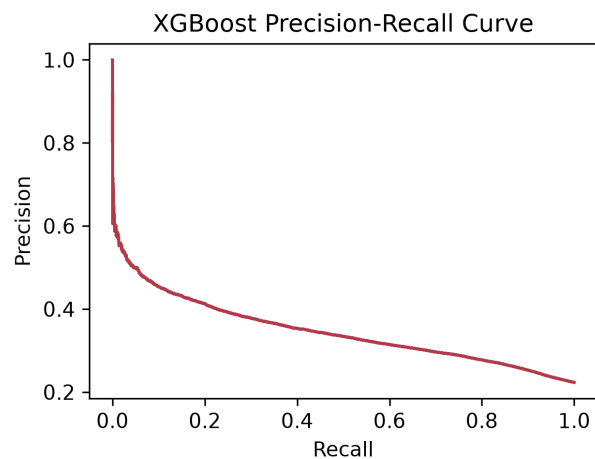
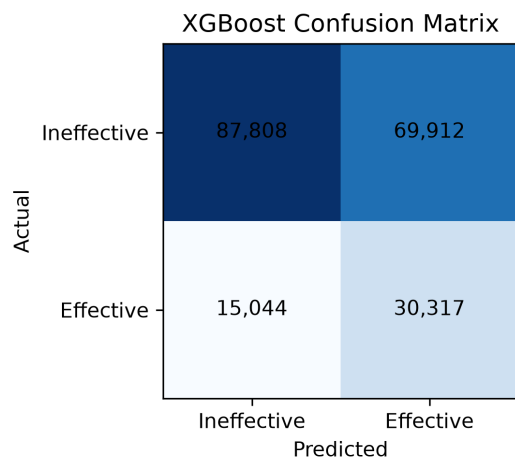
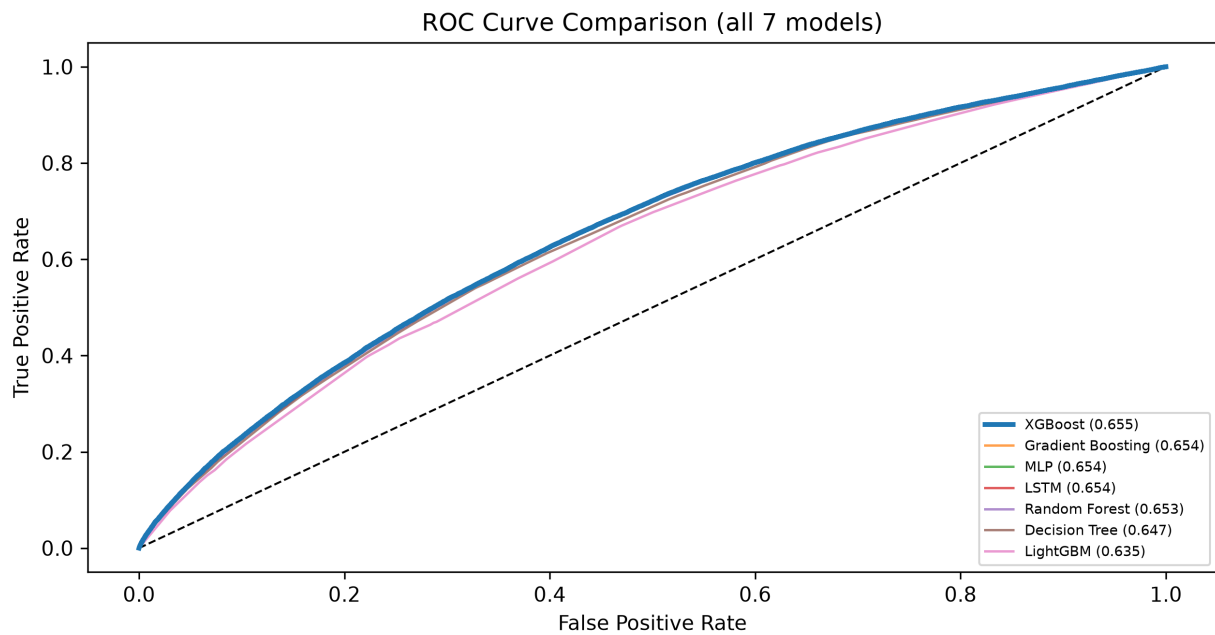
All Models Compared (Test Set)

Rank	Model	PR-AUC	F1	Precision	Recall	ROC-AUC	Accuracy	Diagnosis
1	XGBoost ★	0.3465	0.4165	0.3025	0.6683	0.6545	0.5817	Well-Balanced
2	Gradient Boosting	0.3464	0.4168	0.3043	0.6609	0.6545	0.5868	Well-Balanced
3	MLP	0.3454	0.4174	0.2946	0.7158	0.6540	0.5537	Well-Balanced
4	LSTM	0.3452	0.4162	0.2990	0.6842	0.6540	0.5712	Well-Balanced
5	Random Forest	0.3429	0.4160	0.2986	0.6852	0.6526	0.5702	Moderate Overfit
6	Decision Tree	0.3320	0.4118	0.2878	0.7234	0.6473	0.5385	Well-Balanced
7	LightGBM	0.3164	0.4059	0.2865	0.6955	0.6345	0.5451	Well-Balanced

Recommended for API Deployment: XGBoost ★

XGBoost is deployed behind the project's Flask REST API (**/predict** endpoint, see app.py) and leads the comparison on PR-AUC (0.3465), the primary metric for this imbalanced problem. Its closest competitor, Gradient Boosting, trails by only 0.0002 PR-AUC (effectively tied, likely within run-to-run noise), but XGBoost was chosen for deployment because it combines that top-tier accuracy with well-balanced generalization (train-test PR-AUC gap of 0.0034), fast CPU-only inference (tree_method='hist'), and native early stopping/regularization suited to the 8 GB RAM deployment target. LightGBM was also evaluated as a lighter-weight alternative but under-trained during tuning (early stopping halted after just 4 boosting rounds) and scored lowest of the gradient-boosted models on this run.





Key Takeaways

- All 7 models cluster within a narrow PR-AUC band (0.316-0.347), well above the majority-class floor of ~0.223, but no single model dominates — suggesting the available features carry a real but limited amount of predictive signal.
- Class imbalance (~22.3% Effective) was handled via `scale_pos_weight` rather than resampling, and PR-AUC (not accuracy) was used throughout for tuning/comparison since a trivial always-Ineffective classifier would score ~78% accuracy while catching zero Effective cases.
- XGBoost's confusion matrix (above) favors recall over precision at its tuned threshold: it catches 67% of Effective cases (recall) at the cost of some false positives, a deliberate trade-off for a clinical screening use case where missing a true Effective response is costlier than a false alarm.
- LightGBM under-trained during this run (early stopping halted at only 4 boosting rounds) and is a candidate for a follow-up tuning pass rather than a like-for-like comparison against the other six models.